Tutorial 15

1. How many atoms of helium gas fill a spherical balloon of diameter 30 cm at 20^0C and 1.00 atm? What is the average kinetic energy of the helium atoms?

2. To a first approximation, the Sun consists of a gas of equal numbers of protons and electrons. The temperature at the center of the Sun is about 1×10^7 K, and the density of the Sun is about 1×10^5 kg/m^3. Because the temperature is so high, the protons and electrons are separate particles (rather than being joined together to form hydrogen atoms. Estimate the pressure at the center of the Sun.

3. If a mixture of n_1 moles of monatomic gas and n_2 moles of diatomic gas has γ = 1.5, then what will be the ratio of n_1/n_2.

4. A vessel of volume V = 30 lit contains ideal gas at a temperature of 0 ^{0}C. After a portion of the gas has been let out, the pressure in the vessel is decreased by ΔP = 0.78 atm (temperature remaining constant). Find the mass of the released gas. The gas density under the normal conditions ρ = 1.3 g/lit.